Algorithms 2

Lovasz Local Lemma

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1. Lovasz Local Lemma

1.1 Is it possible to handle dependent bad events?

Definition 1 The scenario:
$E_1, \ldots, E_n$ bad events (over the same sample space).
We wish to prove: $P\left[\bigcap_{i=1}^{n} \overline{E_i}\right] > 0$ – no bad event occurs.

- If $E_1, \ldots, E_n$ are pairwise independent, then we can argue as follows:
  - Recall: events A and B are independent if $P[A \cap B] = P[A]P[B]$.
  - If A, B are independent, then so are $\overline{A}, \overline{B}$:

$$\boxed{\begin{aligned} &P\left[\overline{A} \cap \overline{B}\right] = 1 - P[A \cup B] \\ &= 1 - P[A] - P[B] + P[A \cap B] \\ &= (1 - P[A])(1 - P[B]) \\ &= P\left[\overline{A}\right] P\left[\overline{B}\right] \end{aligned}}$$

1.2 Pairwise independence is not enough

- Back to our problem: $E_1, \ldots, E_n$ pairwise independent $\Rightarrow \overline{E_1}, \ldots, \overline{E_n}$ pairwise independent.
- If, in addition, $P[E_i] < 1$ for all $i \in [n]$, then:

$$P\left[\bigcap_{i=1}^{n} \overline{E_i}\right] = \prod_{i=1}^{n} P\left[\overline{E_i}\right] = \prod_{i=1}^{n} (1 - P[E_i]) > 0$$

  (assuming this equality holds).
- But pairwise independence does **not** imply

$$P\left[\bigcap_{i=1}^{n} \overline{E_i}\right] = \prod_{i=1}^{n} P\left[\overline{E_i}\right].$$

- Mutual independence does.
- What if $E_1, \ldots, E_n$ do depend on one another in some way? The Lovasz local lemma provides a partial solution.

1.3 Mutual independence of random variables

Definition 2 Events $E_1, \ldots, E_n$ are said to be **mutually independent** if

$$P\left[\bigcap_{i \in I} E_i\right] = \prod_{i \in I} P[E_i]$$

whenever $\emptyset \neq I \subseteq [n]$.

Put another way: for every $i \in [n]$ and every $\emptyset \neq I \subseteq [n] \setminus \{i\}$,

$$P\left[E_i \mid \bigcap_{j \in I} E_j\right] = P[E_i].$$

Event A is said to be mutually independent of events $E_1, \ldots, E_n$ if

$$P\left[A \mid \bigcap_{i \in I} E_i\right] = P[A]$$

for all $\emptyset \neq I \subseteq [n]$.

1.4 An equivalent definition

Definition 3 Event A is mutually independent of events $E_1, ..., E_n$ if

$$P\left[A \mid \bigcap_{i=1}^{n} B_i\right] = P[A]$$

where $B_i \in \left\{E_i, \overline{E_i}\right\}$ for all $i \in [n]$.

- No matter whether each E_i occurs or not, this does not affect $P[A]$.
- For a proof of equivalence, see the booklet.

1.5 Pairwise vs mutual independence (example 1)

- Mutual independence $\Rightarrow$ pairwise independence.
- The converse implication fails.

Example. Let $\Omega := \{1, 2, 3, 4\}$ with the discrete uniform measure. $A := \{1, 2\}$, $B := \{1, 3\}$, $C := \{2, 3\}$.

- $P[A] = P[B] = P[C] = \frac{1}{2}$.
- A, B are independent: $P[A \cap B] = P[\{1\}] = \frac{1}{4} = P[A]P[B]$.
- The same applies to B, C and A, C.
- Yet $P[A \cap B \cap C] = P[\emptyset] = 0 \neq P[A]P[B]P[C]$.
- So A, B, C are pairwise independent but **not** mutually independent.

1.6 Pairwise vs mutual independence (example 2)

Example. Two fair dice are rolled. $A := \{\text{sum of outcomes is } 7\}$, $B := \{\text{1st die rolled } 3\}$, $C := \{\text{2nd die rolled } 4\}$.

- $P[A] = P[B] = P[C] = \frac{1}{6}$.
- Events B and C are independent (independent trials).
- To see A and B are independent, note: $P[A \mid B] = P[\text{sum} = 7 \mid \text{1st die} = 3] = P[\text{2nd die} = 4] = \frac{1}{6} = P[A]$.
- Similarly, A and C are independent.
- But $B \cap C \Rightarrow A$, so $P[A \cap B \cap C] = P[B \cap C] = \left(\frac{1}{6}\right)^2$, while $P[A]P[B]P[C] = \left(\frac{1}{6}\right)^3$.

1.7 Complement independence

- Mutual independence is a **stronger** form of independence than pairwise independence.
- We have seen that $E_1, ..., E_n$ pairwise independent $\Rightarrow \overline{E_1}, ..., \overline{E_n}$ pairwise independent.
- This property is also supported in mutual independence:

Theorem 1 If $E_1, ..., E_n$ are mutually independent, then so are $\overline{E_1}, ..., \overline{E_n}$.

For a proof, see the booklet.

1.8 The mutual independence principal

- The following is a handy tool for determining mutual independence.

Theorem 2 (Mutual independence principal)
Let $X := (X_1, ..., X_m)$ be a sequence of pairwise independent trials. Let $A_1, ..., A_n$ be events, and for each $i \in [n]$ let A_i be determined by trials $F_i \subseteq X$. Given $I \subseteq [n]$ and $j \in [n] \setminus I$, if

$$F_j \cap \left(\bigcup_{i \in I} F_i\right) = \emptyset,$$

then A_j is mutually independent of $(A_i)_{\{i \in I\}}$.

1.9 Statement of the Lovasz local lemma

- There are several versions. We state one of those.
- The version stated here is called the **symmetric version**.
- We focus on applications of the lemma and not its proof(s).

The dependency graph

Definition 4 Events $E_1, \ldots, E_n$. A dependency graph $D := D(E_1, \ldots, E_n)$ is any graph whose vertex set is $E_1, \ldots, E_n$ and supporting the property: for every $i \in [n]$, event E_i is mutually independent of the events $\{E_j : j \neq i \wedge \{i, j\} \notin E(D)\}$.

Remark

Sometimes a directed version of such graphs is more useful (i.e., in **asymmetric** applications).

1.9 Statement of the Lovasz local lemma

Remark

Danger: We did **not** just say that we build a graph on $E_1, \ldots, E_n$ by placing an edge (E_i, E_j) if E_i, E_j are dependent.

1.10 The symmetric local lemma

Theorem 3 (The symmetric local lemma)
Let $E_1, ..., E_n$ be events such that the following holds:

1. **(Symmetry)** $P[E_i] \leq p$ for all $i \in [n]$.
2. **(Limited dependency)** $\Delta(D(E_1, ..., E_n)) \leq d$.
3. **(Bound p and d)** $e \cdot p \cdot (d + 1) \leq 1$ (up to a constant).

Then $P\left[\bigcap_{i=1}^{n} \overline{E_i}\right] > 0$.